CSCI E-106:Assignment 2

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Due Date: February 13, 2026 at 11:59 pm EST

Problem 1

Refer to the CDI data set. Use the number of active physicians as Y and total personal income as X.

```r
cdi <- read.csv("CDI Data.csv")
cdi = cdi[,c("Total.personal.income", "Number.of.active.physicians")]
colnames(cdi) = c("TPI", "Active.Physicians")
```

a-) Build a regression model to predict Y by using X. Write down the standard error of the slope and intercept.

```r
fit = lm(Active.Physicians ~ TPI, cdi)
summary(fit)
```

```
##
## Call:
## lm(formula = Active.Physicians ~ TPI, data = cdi)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1926.6  -194.5   -66.6    44.2  3819.0
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -48.39485   31.83333   -1.52    0.129
## TPI           0.13170    0.00211   62.41   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 569.7 on 438 degrees of freedom
## Multiple R-squared:  0.8989, Adjusted R-squared:  0.8987
## F-statistic:  3895 on 1 and 438 DF,  p-value: < 2.2e-16
```

The standard error of the slope is 0.00211.

The standard error of the intercept is 31.83333.

b-) Are the slope and intercept statistically significant? Write down your null and alternative hypothesis and p value of the test. Calculate the 95% Confidence interval for the slope.

The null hypothesis for the intercept is that the intercept is 0. The alternative hypothesis for the intercept is that the intercept is not 0.

The t-statistic for the intercept at -48.39485 is -1.52. The probability of getting a t-statistic at least as large as -1.52 under the null hypothesis is 0.129.

If I take a significance level to be 0.05, I cannot discharge the null hypothesis, because $0.129 > 0.05$.

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The null hypothesis for the slope is that the slope is 0. The alternative hypothesis for the slope is that the slope is not 0.

The t-statistic for the slope at 0.13170 is 62.41. The probability of getting a t-statistic at least as large as 62.41 is $2 * 10^{-16}$.

If I take a significance level to be 0.05, I can discharge the null hypothesis, because $2 * 10^{-16} < 0.05$.

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The formula for the confidence interval of the coefficient of a linear regression is:

$$CI = \beta \pm t^{*} SE_{\beta}$$

where β is the coefficient, t^{*} is the critical value, and SE_{β} is the standard error of the coefficient.

In this case, β is 0.13170, the critical value is 62.41, and the standard error is 0.00211.

```
coefficients = summary(fit)$coefficients["TPI",]

estimate = coefficients[["Estimate"]]
standard_error = coefficients[["Std. Error"]]
degrees_of_freedom = df.residual(fit)

critical_value = qt(.095/2., degrees_of_freedom)

margin_of_error = critical_value * standard_error

lower_bound = estimate - margin_of_error
upper_bound = estimate + margin_of_error
```

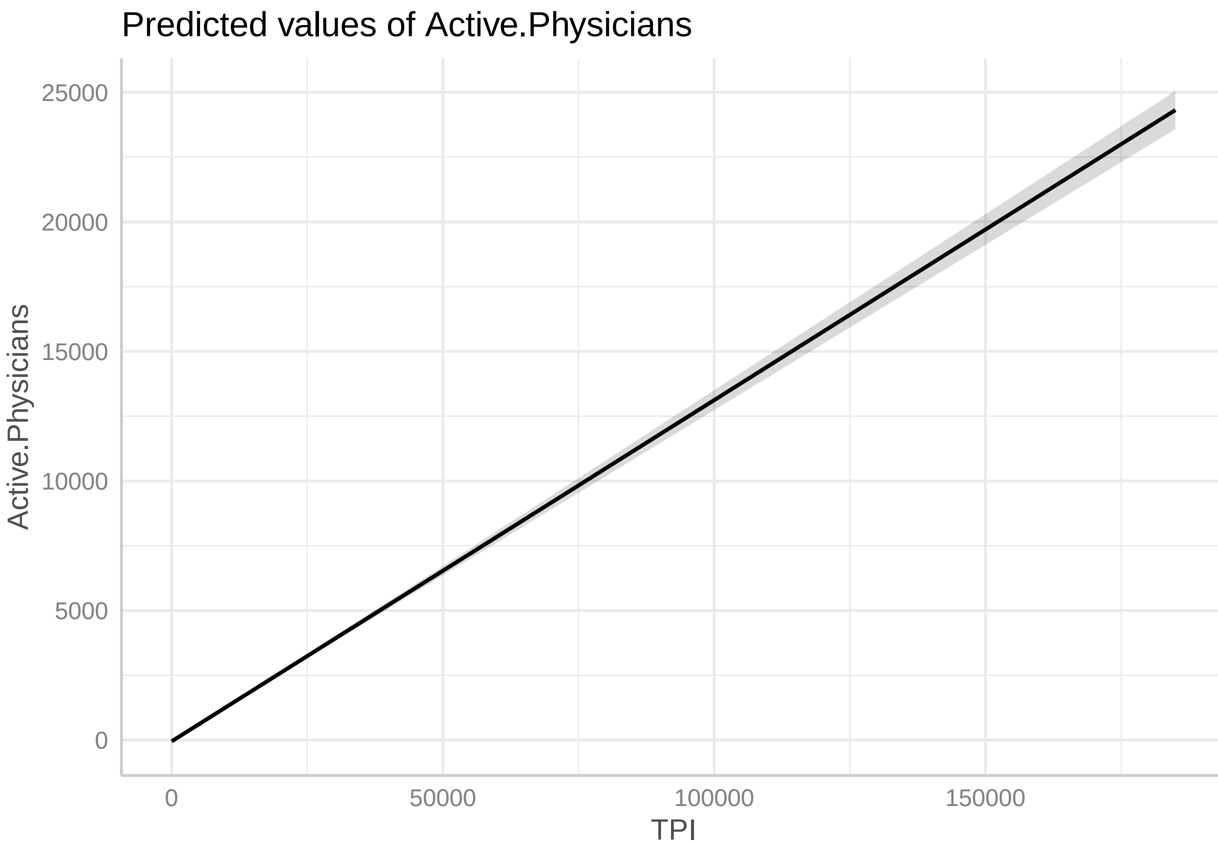
The confidence interval for the slope is (0.1352321, 0.1281702).

c-) Predict the number of active physicians for total personal income of 100,000, 250,000 and 50,000. Calculate the 95% confidence interval (Hint: Use the range of total personal income to identify which observations are within the range of the training data)

```
predictions = data.frame(TPI=c(100000, 250000, 50000))
predictions$Active.Physicians = predict(
  fit, newdata = predictions, interval="confidence", level=0.95
)
print(predictions)
```

```
##      TPI Active.Physicians.fit Active.Physicians.lwr Active.Physicians.upr
## 1 100000             13121.724             12735.899             13507.549
## 2 250000             32876.902             31871.240             33882.565
## 3  50000              6536.665              6353.955              6719.374
```

```
pred <- ggpredict(fit, terms = "TPI", ci.lvl = 0.95)
plot(pred)
```



The 95% confidence interval for active physicians where TPI = \$100,000 is $(12735.899, 13507.549)$.

The 95% confidence interval for active physicians where TPI = \$250,000 is $(31871.240, 33882.565)$.

The 95% confidence interval for active physicians where TPI = \$50,000 is $(6353.955, 6719.374)$.

d-) What is the error variance of the regression line?

The formula for error variance of a regression line is

$$S^2 = \frac{SSE}{n-2}$$

The SSE and (n-2) are 569.7 and 438 respectively. This gives $S^2 = 1.300$.

Problem 2

Refer to the CDI data set. Use the number of active physicians as Y and total personal income as X. Select 1,000 random samples of 400 observations, fit the regression model and record β_0 and β_1 for each selected sample. Calculate the mean and variance of β_0 and β_1 based on the 1,000 different regression line and compare against the regression model in question 1 part b. (50 points)

```
data = data.frame()
for (i in 1:1000) {
  sample = cdi[sample(nrow(cdi), 400), ]
  fit = lm(Active.Physicians ~ TPI, sample)
  data = rbind(data, coef(fit))
}
stats = data.frame(
  mean=c(mean(data[,1]), mean(data[,2])),
```

```r
  var=c(var(data[,1]), var(data[,2])),
  stderr=c(sd(data[,1]), sd(data[,2]))
)
rownames(stats) = c("(Intercept)", "TPI")
print(stats)
```

```
##                    mean          var      stderr
## (Intercept) -48.949032 7.219514e+01 8.496772297
## TPI           0.131774 1.089055e-06 0.001043578
```

The estimates of mean are pretty much the same. The variances/stderrs of the intercepts differ by a factor of ~5.

The difference in variances of the intercepts may be less important because the intercept already fails to differ significantly from 0.